CAPA: Complementary Adaptive Phase Attention for Periodic Time-Series Forecasting

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Abstract. Long-term time-series forecasting is often dominated by periodic structure, yet periodicity is rarely homogeneous inside a cycle. Different phase positions may be aligned, offset, or complementary across variables and samples. Existing attention forecasters usually emphasize high similarity and therefore capture phase-aligned repetition more naturally than offset phase candidates. We propose CAPA, a phase-view forecasting model for periodic heterogeneity. The model first rearranges a history window by a specified period length, so that each token represents a phase position inside the period rather than an ordinary time index. It then introduces a positive-negative phase attention module with two branches: a positive branch for phase-aligned dependencies and a negative branch for complementary offset dependencies. A dynamic fusion gate adapts the contribution of the two branches for each input window. Experiments on seven long-term forecasting benchmarks show that CAPA is competitive with phase-based, TimeXer, TQNet, PatchTST, and iTransformer. Direct-test ablations using the same experimental protocol further support that positive phase attention, negative phase attention, and dynamic fusion each contribute to forecast accuracy.

Keywords: Long-term forecasting · Periodic heterogeneity · Phase attention · Time series forecasting.

1 Introduction

Periodic behavior is a central regularity in many multivariate time series, including electricity load, traffic flow, meteorology, and exchange or sensor measurements. A daily load curve, for example, is not only a sequence of neighboring points. It also contains repeated phase positions: morning peaks, midday valleys, evening peaks, and night-time lows. These positions may repeat from period to period, but they do not all interact in the same way. Some phases are positively aligned and co-vary across cycles, whereas other phases exhibit stable offset or complementary relations.

Most sequence forecasters model the history as an ordinary chain of time tokens. This representation is flexible, but it can make the model spend capacity rediscovering the periodic layout. Patch and period-aware models partly address the issue by grouping time points or using periodic priors. However, many attention mechanisms still interpret high similarity as strong dependence. As a result,

they naturally emphasize co-moving phase positions and may under-represent offset phase candidates such as peak–valley alternation, day–night compensation, or directional shifts between different parts of a cycle.

This paper studies periodic dependence under aligned and offset phase relations. We instantiate this idea with CAPA, a complementary adaptive phase attention forecaster. Instead of treating the history only as a raw time sequence, CAPA builds a phase token space and learns two explicitly different relational views over it. If the history length is L and the period length is P, the sequence is reshaped into periods, and the model attends over the P phase positions. Each phase token summarizes its observations across historical periods. This design directly matches the periodic forecasting problem: the model identifies which positions inside a period support future values and gives aligned and offset phase candidates separate parameterizations.

The main contributions are summarized as follows:

- We formulate periodic heterogeneity as a two-view phase-dependency problem, where aligned and offset phases are treated as different candidate sources of predictive evidence inside the same cycle.
- We design a positive-negative phase attention module as the core operator. The positive branch captures phase-aligned periodic patterns; the negative branch captures complementary offset phase interactions.
- We introduce a dynamic fusion gate that adaptively combines positive and negative phase outputs for each sample rather than using a fixed multi-branch weight.
- We evaluate CAPA on seven standard forecasting datasets and compare it with representative phase-based and transformer baselines. Component ablations on representative datasets use the same experimental protocol and change only one component at a time.

2 Periodic Heterogeneity: Motivation

The central observation behind CAPA is that periodicity in real multivariate sequences is structured but not homogeneous. A period is often used as a single prior: one day, one week, one traffic cycle, or one seasonal cycle. However, after a sequence is folded into that period, the different positions inside the cycle play different roles. Some phases are aligned with the same semantic state, such as the morning rise of electricity demand across consecutive days. Other phases are offset but still informative, such as a night valley that constrains the following daytime peak. Treating all useful dependencies as similarity-based interactions can therefore hide part of the periodic signal.

Definition 1 (Phase token). *Given a history window $X \in \mathbb{R}^{L\times C}$ and a period length P, the p-th phase token of variable c is the vector $\{X_{p+kP,c}\}_{k=0}^{N-1}$, where $N = \lceil L/P \rceil$. It collects the observations that share the same within-period position.*

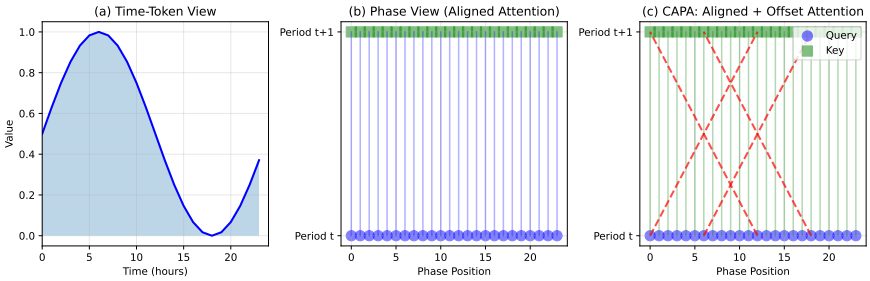


Fig. 1. Motivation of complementary phase modeling. Similarity-only phase attention naturally emphasizes aligned or nearby phases, while periodic forecasting can also benefit from offset phase candidates such as peak–valley or day–night relations. CAPA keeps the aligned view and adds an offset-biased candidate view, then uses a dynamic gate to adapt their contributions.

Definition 2 (Positive and negative phase dependency). *For two phase positions p_i and p_j, a positive phase dependency indicates that their variations are aligned or co-moving after periodic folding. A negative phase dependency denotes a complementary relation between offset phase positions after periodic folding. Both dependencies are learned by branch-specific attention projections and the forecasting objective.*

This distinction is visible in many forecasting scenarios. In electricity data, household and industrial usage can share strong positive daily recurrence around similar working hours, while low-load night phases can predict the amplitude of the next peak. In traffic data, the rush-hour crest and the subsequent valley are not similar in value but jointly describe road capacity and commuting rhythm. In weather data, day-night alternation produces offset patterns in temperature, humidity, and pressure. A model that only amplifies high-similarity phase pairs can use the first type of dependency but may attenuate the second type.

The phase representation also explains the effectiveness of a lightweight model. When the period length P is much smaller than the input length L, attention over phases uses P tokens rather than L raw time tokens. The model no longer needs to recover the periodic lattice from scratch; it can spend its capacity on learning how positions inside the cycle interact. CAPA uses this representation as the substrate for a distinct two-view operator: aligned and offset-biased phase candidates are parameterized separately and then fused adaptively.

3 Related Work

Long-term time-series forecasting. Transformer-style forecasting models have advanced long-horizon prediction by using sparse attention, decomposition, patching, and channel-wise modeling. Informer [1] reduces attention cost for long sequences, Autoformer [3] introduces series decomposition and auto-correlation, and PatchTST [4] uses patch tokens to improve representation

efficiency. Other representative directions include general temporal representation learning such as TimesNet [5] and SCINet [18], efficient attention [8–10], frequency or decomposition modeling [11, 12, 16, 17], simple linear baselines and normalization-aware designs [13, 14], cross-variable or cross-dimension modeling [6, 15], and recent multi-scale or query-based temporal mixing models [19–21]. These methods are general sequence forecasters. CAPA is complementary: it targets periodic heterogeneity by changing the tokenization from time view to phase view.

Periodic and patch-based forecasting. Period-aware methods exploit the observation that future sequences often repeat earlier cycles. PhaseFormer [7] organizes a sequence into periodic patches and models phase-level structure. These methods show the value of periodic representations. CAPA addresses a different missing piece: within a chosen period, it gives aligned and offset-biased phase relations separate attention parameterizations and combines them with a sample-adaptive fusion gate. More broadly, CAPA belongs to the family of period-aware models but focuses on the interaction structure between phases rather than on token construction alone. It studies whether a two-view phase attention block can improve long-horizon forecasting under the standard benchmark protocol.

Attention beyond similarity. Standard dot-product attention treats large query-key similarity as strong affinity. This is effective for aligned patterns, but a low-similarity phase can still be predictive if it carries stable offset evidence for the target phase. Positive-negative phase attention addresses this limitation with two parallel branches. The positive branch uses a closeness-biased phase relation, whereas the negative branch uses an offset-biased phase relation and leaves the final relevance to learned projections and the forecasting loss. The final prediction is controlled by a learned gate.

Efficient and structured forecasting. Many recent works improve long-horizon forecasting by changing the representation on which the backbone operates. Patch-based models reduce token length, period-aware models reorganize temporal structure, graph-based models inject relational priors, and normalization-aware designs such as RevIN [22] improve robustness to distribution shift. CAPA is in this structure-first family: it defines a compact phase token space and designs an attention operator specialized for periodic heterogeneity.

4 Problem Setting and Phase View

Let $X \in \mathbb{R}^{B \times L \times C}$ be a batch of multivariate history windows, where B, L, and C denote batch size, input length, and number of variables. The goal is to predict $Y \in \mathbb{R}^{B \times H \times C}$, where H is the forecast horizon. Given a period length P, CAPA pads X circularly if needed and reshapes it into

$$X_p \in \mathbb{R}^{B \times C \times N \times P},$$

where $N = \lceil L/P \rceil$ is the number of input periods. The tensor is then permuted into a phase view

$$X_\phi \in \mathbb{R}^{B \times C \times P \times N}.$$

Table 1. Notation used in the phase-view formulation.

Symbol	Meaning
L	look-back window length
H	prediction horizon
C	number of variables
P	configured period length
N	number of historical periods after padding
M	number of predicted periods, $M = \lceil H/P \rceil$
d	hidden dimension of each phase token
D_ϕ	circular phase-distance matrix
O^+, O^-	positive and negative phase outputs
G	dynamic fusion gate

For each variable, a phase token contains the historical values observed at the same phase position across periods. A linear embedding maps each token from N historical periods to a latent dimension d, producing $Z \in \mathbb{R}^{(BC)\times P\times d}$.

This representation changes the modeling target. Attention is no longer primarily about adjacent time steps; it is about relations among phase positions in one period. This is useful when the relevant dependency is periodic but not local in the original time axis.

4.1 Forecasting Protocol and Metrics

For a dataset split into training, validation, and test intervals, the model uses a sliding window protocol. Each sample contains a look-back window of length L and a prediction target of length H. In all reported experiments, the input length is fixed to $L = 720$, following the full-result protocol used in the comparison table. We report the standard MSE and MAE metrics, where lower values indicate better forecasting performance. Reporting both metrics is useful because periodic models can reduce phase-shifted large errors without always changing the typical absolute error by the same amount.

4.2 Circular Phase Distance

Because phase positions are cyclic, the distance between phases should also be cyclic. For $p_i, p_j \in \{0, \ldots, P-1\}$, we define

$$D_\phi(i,j) = \min(|p_i - p_j|, P - |p_i - p_j|).$$

This distance is small for adjacent phases and also small for positions close to the period boundary. The positive branch uses D_ϕ as a closeness prior, whereas the negative branch reverses the role of the distance bias so that offset phases can receive attention. This design prevents the model from treating the first and last phase as unrelated only because they are far apart in the unfolded index.

5 The CAPA Framework

Figure 2 illustrates the overall architecture. The model contains four main stages: reversible normalization, phase-view embedding, positive-negative phase attention, and phase prediction.

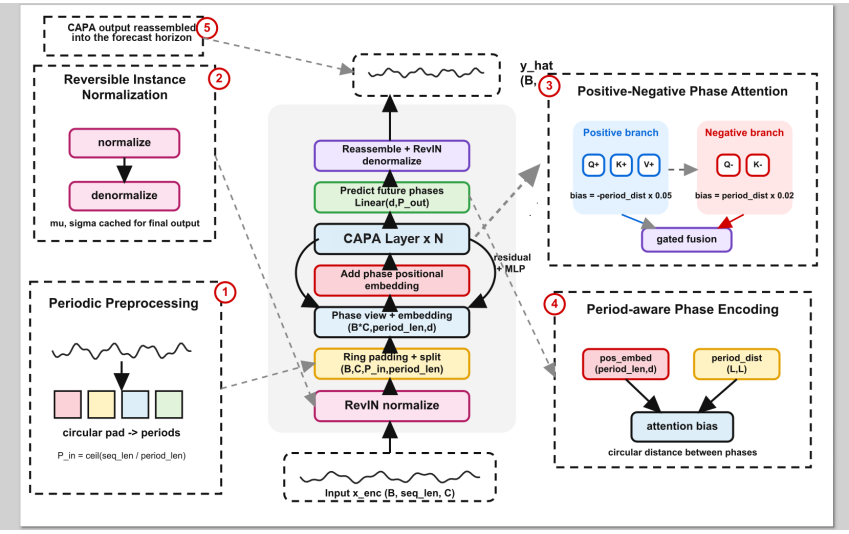


Fig. 2. Overview of CAPA. The input is normalized, circularly padded, and reshaped into a phase view. Positive and negative phase attention branches model aligned and offset-biased candidate relations, and a dynamic gate fuses them before the phase predictor reconstructs the forecast horizon.

5.1 Reversible Normalization and Phase Embedding

The model applies RevIN normalization to reduce distribution shift:

$$\mu, \sigma \in \mathbb{R}^{B\times 1\times C}, \qquad \tilde{X} = (X-\mu)/\sigma \in \mathbb{R}^{B\times L\times C},$$

where μ and σ are computed along the temporal dimension for each sample and variable. After forecasting, the prediction is mapped back through the inverse transformation. The normalized sequence is circularly padded so that its length is divisible by P:

$$\tilde{X}_{\text{pad}} \in \mathbb{R}^{B\times NP\times C}, \qquad X_\phi = \text{permute}(\text{reshape}(\tilde{X}_{\text{pad}})) \in \mathbb{R}^{B\times C\times P\times N}.$$

We flatten the batch and variable dimensions as $X_\phi^\flat \in \mathbb{R}^{(BC)\times P\times N}$. A linear embedding maps each phase-across-period vector to a latent representation:

$$Z = \text{LN}(X_\phi^\flat W_e + b_e) + E_\phi \in \mathbb{R}^{(BC)\times P\times d},$$

where $W_e \in \mathbb{R}^{N\times d}$, $b_e \in \mathbb{R}^d$, and $E_\phi \in \mathbb{R}^{P\times d}$ is a learnable phase positional embedding broadcast over BC.

5.2 Positive Phase Attention

The positive branch models phase-aligned dependencies. For phase tokens Z, the branch computes multi-head projections

$$Q^+, K^+, V^+ \in \mathbb{R}^{(BC)\times h\times P\times d_h},$$

where h is the number of heads and $d = hd_h$. The attention score is

$$A^+ = \mathrm{softmax}\left(\frac{Q^+(K^+)^\top}{\sqrt{d_h}} - \lambda_+ D_\phi\right),$$

where $D_\phi \in \mathbb{R}^{P\times P}$ is broadcast to the head dimension and $A^+ \in \mathbb{R}^{(BC)\times h\times P\times P}$. The negative sign encourages closer phase positions to receive stronger positive attention. After head concatenation, the output is $O^+ \in \mathbb{R}^{(BC)\times P\times d}$.

5.3 Offset-Biased Negative Phase Attention

The negative branch targets candidate phase-offset relations. It uses separate query and key projections:

$$Q^-, K^- \in \mathbb{R}^{(BC)\times h\times P\times d_h}.$$

The score is

$$A^- = \mathrm{softmax}\left(\frac{Q^-(K^-)^\top}{\sqrt{d_h}} + \lambda_- D_\phi\right).$$

The positive distance bias increases the relative influence of phase-offset positions. Here $A^- \in \mathbb{R}^{(BC)\times h\times P\times P}$. The implementation shares the value projection with the positive branch, yielding $O^- = A^- V^+ \in \mathbb{R}^{(BC)\times P\times d}$ after head concatenation. The learned query/key projections and the forecasting objective select useful offset phases for each dataset and horizon.

5.4 Dynamic Fusion Gate

Positive and negative dependencies do not have fixed importance. Some samples are dominated by repeated aligned cycles, while others show peak–valley or offset relations. CAPA therefore learns a gate

$$G = \sigma(W_g\,\mathrm{mean}_\phi(Z) + b_g) \in \mathbb{R}^{(BC)\times 1\times d},$$

and fuses the branches as

$$O = G \odot O^+ + (1 - G) \odot O^- \in \mathbb{R}^{(BC)\times P\times d}.$$

The fused representation is projected and added residually:

$$Z' = Z + \mathrm{Dropout}(OW_o) \in \mathbb{R}^{(BC)\times P\times d}.$$

An MLP block and layer normalization further refine the phase representation.

Table 2. Conceptual complexity comparison for one attention block. Constants, heads, and projection dimensions are omitted for clarity.

Modeling view	Tokens	Attention cost
Time-token attention	L	$O(L^2)$
Patch-token attention	L/S	$O((L/S)^2)$
Phase-token attention	P	$O(P^2)$
CAPA	P	$O(2P^2)$

5.5 Complexity and Parameterization

The computational advantage of the phase view comes from attending over within-period positions. If a standard attention layer uses L time tokens, its attention matrix has size $L \times L$. In CAPA, the attention matrix has size $P \times P$, while information across historical periods is compressed into the phase embedding. When $P \ll L$, this substantially reduces the attention cost. The positive-negative design doubles the score computation but keeps the same phase-token length, so the model remains compact.

Remark 1. The factor two in CAPA corresponds to the positive and negative branches, not to a longer token sequence.

5.6 Prediction Head

After the gated phase attention block, the model maps refined phase representations to future coefficients. Let $M = \lceil H/P \rceil$ denote the number of output periods needed to cover the forecasting horizon. The prediction head is a linear map along the hidden dimension:

$$U = Z' W_o^{\text{head}} + b_o \in \mathbb{R}^{(BC)\times P\times M},$$

where $W_o^{\text{head}} \in \mathbb{R}^{d\times M}$ and $b_o \in \mathbb{R}^M$. We then restore the batch and variable dimensions, permute the period and phase axes, and unfold the sequence:

$$U_r = \text{reshape}(U) \in \mathbb{R}^{B\times C\times P\times M},$$

$$\tilde{Y}_{\text{pad}} = \text{flatten}_{(M,P)}\left(\text{permute}(U_r)\right) \in \mathbb{R}^{B\times MP\times C}.$$

The normalized forecast keeps the first H time steps with zero-based slicing,

$$\hat{\tilde{Y}} = \tilde{Y}_{\text{pad}}[:, : H, :] \in \mathbb{R}^{B\times H\times C},$$

and the inverse RevIN transform restores the sample-level mean and scale:

$$\hat{Y} = \hat{\tilde{Y}} \odot \sigma + \mu \in \mathbb{R}^{B\times H\times C}.$$

Algorithm 1 Forward pass of CAPA

Require: history window X, period length P
1: Normalize X with RevIN.
2: Circularly pad and reshape X into phase view X_ϕ.
3: Embed phase-across-period tokens and add phase positional embeddings.
4: Compute positive phase attention O^+ with closeness-biased scores.
5: Compute negative phase attention O^- with offset-biased scores.
6: Compute gate G from the global phase representation.
7: Fuse $O = G \odot O^+ + (1 - G) \odot O^-$.
8: Predict future period coefficients and reshape them to horizon H.
9: Apply inverse RevIN and return the forecast.

Table 3. Dataset statistics used in experiments. T denotes the look-back length and H denotes the forecasting horizon range.

Dataset	Var	Length	T	H	Freq	Scale
ETTh1	7	14,400	720	96–720	1 hour	0.1M
ETTh2	7	14,400	720	96–720	1 hour	0.1M
ETTm1	7	57,600	720	96–720	15 mins	0.4M
ETTm2	7	57,600	720	96–720	15 mins	0.4M
Weather	21	52,696	720	96–720	10 mins	1.1M
Electricity	321	26,304	720	96–720	1 hour	8.1M
Traffic	862	17,544	720	96–720	1 hour	15.0M

5.7 Need for Dynamic Fusion

A fixed fusion rule assumes that the relative importance of aligned and offset relations is constant across all windows. This assumption is too restrictive for long-term forecasting. The same dataset may contain weekdays and weekends, stable and transitional weather segments, or congested and free-flow traffic intervals. The gate in CAPA is conditioned on the global phase representation, so its value changes with the current sample. In windows with regular repetition, the gate can emphasize the positive branch. In windows where the current phase is better explained by offset phase candidates, it can increase the contribution of the negative branch.

6 Experiments

6.1 Datasets and Baselines

We evaluate on seven standard long-term forecasting datasets: ETTh1, ETTh2, ETTm1, ETTm2, Weather, Electricity, and Traffic. Each dataset is evaluated at horizons $H \in \{96, 192, 336, 720\}$. The compared models are PhaseFormer, TimeXer [20], TQNet [21], PatchTST [4], and iTransformer [6]. MSE and MAE are used as the evaluation metrics, and lower values indicate better forecasting performance.

Implementation details. Table 3 reports the dataset statistics used in our experiments. The data loading and preprocessing procedures follow the standard long-term forecasting protocol used in prior works: chronological

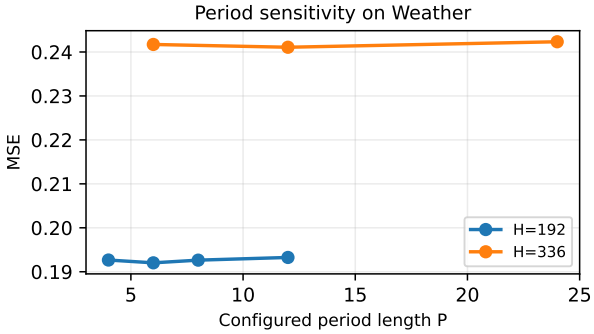


Fig. 3. CAPA period sensitivity on Weather. The results are obtained from experiments with different period lengths; lower MSE indicates better forecasting accuracy in this figure.

train/validation/test splits are kept unchanged, each model uses a look-back window of 720 time steps, and the prediction horizons are $\{96, 192, 336, 720\}$. All reported forecasts are produced by direct multi-step prediction rather than recursive one-step rollout.

All baseline methods are implemented based on their original papers or official code. For cases where fixed random seeds are not specified, each experiment is repeated three times to ensure stability. All experiments are conducted using PyTorch [2] on a single NVIDIA A100 24GB GPU. Model selection is based on the validation loss, and the final tables report MSE and MAE on the test split.

For CAPA, unless a dataset-specific setting is listed in Table 4, all experiments use label length 48, Adam optimization, batch size 32, the `type1` learning-rate schedule, RevIN without affine parameters ($\epsilon = 10^{-5}$), and Huber loss with $\delta = 1.0$. All baseline models are trained with the same Huber loss setting to keep the optimization objective consistent across comparisons. The ablation section uses the trained full CAPA model as the reference and changes only one component at a time.

Period setting. For model configuration, the primary period is obtained from frequency-domain analysis by identifying the dominant spectral component and remains fixed throughout training. Following this phase-length selection procedure, CAPA uses $P = 24$ for ETTh, Weather, Electricity, and Traffic, and $P = 15$ for the 15-minute ETTm datasets. To assess sensitivity to the selected phase length, we conduct a CAPA period-sensitivity experiment on Weather with multiple candidate periods. As shown in Fig. 3, the tested periods give close but distinguishable errors, demonstrating that CAPA captures the dataset's primary periodic structure.

Experimental results. The results of CAPA, PhaseFormer, TQNet, TimeXer, and iTransformer are obtained from experiments under the same comparison protocol. PatchTST values follow the same comparison protocol used in Table 5. For cases where fixed random seeds are not specified, each experiment is repeated three times to ensure stability.

Table 4. Training and CAPA hyperparameters used for the reported experiments.

Group	Setting	Value
Forecasting protocol	look-back / label / horizon	720/48/{96, 192, 336, 720}
Training	optimizer / learning rate schedule	Adam / `type1`
Training	batch size / workers	32 / 10
Training	maximum epochs / early stopping	20 / validation loss
Training	loss / Huber delta	Huber / 1.0
Runtime	device / mixed precision	single GPU `cuda:0` / disabled
CAPA phase view	period length P	24 for ETTh, Weather, Electricity, Traffic; 15 for ETTm
CAPA embedding	latent dimension	8
CAPA attention	layers / heads / dropout	1 / 4 / 0.1
CAPA MLPs	phase encoder / predictor hidden size	32 / 64
Normalization	RevIN / affine / ϵ	enabled / disabled / 10^{-5}

6.2 Main Results

Table 5 reports the full comparison. CAPA is strong across the majority of dataset–horizon pairs. The results suggest that explicitly modeling an offset-biased phase view can be useful across ETT, Weather, Electricity, and Traffic benchmarks, especially on longer horizons where phase-offset candidates may affect the forecast. Compared with phase-only and general transformer baselines, CAPA provides a dedicated mechanism for testing offset phase evidence rather than treating all dependencies as similarity-driven interactions.

Table 5. Full results across datasets and prediction lengths. Each entry reports MSE and MAE. The input length is set to 720. The best results are marked with **Bold**, and the second-best results are marked with Underlined.

Dataset	Horizon	CAPA		PhaseFormer		TQNet		TimeXer		iTransformer		PatchTST	
		ours		2026		2025		2024		2024		2023	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.358**	**0.386**	0.366	0.388	0.371	0.393	0.441	0.444	0.389	0.421	0.377	0.408
	192	**0.395**	**0.408**	0.401	0.409	0.427	0.423	0.471	0.468	0.424	0.446	0.413	0.431
	336	**0.421**	**0.425**	0.432	0.432	0.475	0.446	0.495	0.494	0.456	0.469	0.436	0.444
	720	**0.425**	**0.446**	0.436	0.449	0.489	0.470	0.507	0.517	0.545	0.532	0.455	0.475
ETTh2	96	0.278	0.341	0.281	0.345	0.286	0.350	0.292	0.352	0.305	0.361	**0.276**	**0.339**
	192	**0.336**	**0.378**	0.342	0.382	0.348	0.390	0.345	0.393	0.405	0.421	0.342	0.385
	336	**0.364**	**0.404**	0.366	0.406	0.392	0.420	0.393	0.423	0.411	0.436	**0.364**	0.405
	720	0.397	**0.434**	0.398	0.436	0.414	0.447	0.419	0.454	0.448	0.470	**0.395**	**0.434**
ETTm1	96	0.302	**0.350**	0.338	0.375	0.299	0.351	0.327	0.363	0.315	0.369	**0.298**	0.352
	192	0.337	**0.369**	0.360	0.395	0.340	0.376	0.351	0.385	0.349	0.388	**0.335**	0.373
	336	**0.365**	**0.385**	0.383	0.407	0.367	0.391	0.377	0.395	0.381	0.409	0.366	0.389
	720	**0.420**	**0.413**	0.433	0.435	0.438	0.433	0.440	0.442	0.437	0.439	**0.420**	0.421
ETTm2	96	0.175	0.266	0.182	0.273	0.173	0.264	0.181	0.272	0.179	0.274	**0.165**	**0.260**
	192	0.231	0.303	0.236	0.307	0.230	0.301	0.235	0.310	0.239	0.314	**0.219**	**0.298**
	336	0.280	0.334	0.282	0.338	0.284	0.340	0.292	0.343	0.309	0.356	**0.268**	**0.333**
	720	**0.348**	**0.377**	0.356	0.384	0.371	0.398	0.379	0.406	0.387	0.407	0.352	0.386
Weather	96	**0.149**	**0.196**	0.152	0.201	0.152	0.205	0.160	0.212	0.159	0.212	**0.149**	0.199
	192	**0.192**	**0.236**	0.198	0.244	0.207	0.257	0.205	0.255	0.203	0.252	0.193	0.243
	336	0.241	**0.277**	0.248	0.283	0.265	0.300	0.259	0.295	0.253	0.291	**0.240**	0.281
	720	0.313	**0.331**	0.314	0.332	0.326	0.343	0.339	0.350	0.317	0.337	**0.312**	0.334
Electricity	96	0.136	**0.229**	0.136	**0.229**	0.143	0.244	0.152	0.245	**0.135**	0.233	0.141	0.240
	192	0.156	**0.248**	0.158	0.249	**0.155**	0.253	0.157	0.258	**0.155**	0.253	0.156	0.256
	336	0.171	**0.264**	0.172	**0.264**	0.170	0.267	0.178	0.272	**0.169**	0.267	0.172	0.267
	720	**0.203**	**0.290**	0.214	0.299	0.221	0.315	0.213	0.310	0.204	0.301	0.208	0.299
Traffic	96	**0.362**	**0.239**	0.364	0.244	0.394	0.288	0.383	0.275	0.374	0.273	0.363	0.250
	192	**0.377**	**0.246**	0.378	0.255	0.406	0.295	0.392	0.285	0.393	0.283	0.382	0.258
	336	**0.392**	**0.254**	0.393	0.260	0.424	0.305	0.426	0.308	0.409	0.292	0.399	0.268
	720	0.433	**0.276**	0.434	0.279	0.474	0.339	0.464	0.328	0.450	0.314	**0.432**	0.289

Table 6 reports model efficiency under the same experimental setting. CAPA keeps a compact parameter scale while adding the two-view phase attention

Table 6. Model efficiency. Parameters and FLOPs are calculated from the experimental settings, and running time is measured from experiments as the average seconds per training iteration.

Model	Params (M)	FLOPs (G)	Time (s/iter)
CAPA	0.002	0.224	0.093
PhaseFormer	0.002	0.181	0.075
TQNet	0.660	0.412	0.132
TimeXer	7.364	1.086	0.102
iTransformer	7.151	0.964	0.100

operator. Its running time remains lower than the compared general transformer baselines, showing that the added positive-negative phase modeling does not introduce a large efficiency burden.

Selection of representative ablation datasets. The ablation table does not include every dataset because the purpose is to evaluate the three architectural components rather than reproduce the full main comparison. ETTh1 and ETTm2 cover different sampling granularities in the ETT family. Weather tests smooth physical variables with strong day-night structure. Electricity and Traffic test large multivariate systems where offset or peak-valley candidates are common. This subset follows a compact ablation style while still spanning multiple domains.

6.3 Ablation Study

To keep the component analysis compact, we avoid listing every horizon again and instead average the results over horizons $\{96, 192, 336, 720\}$. We use five representative datasets: ETTh1, ETTm2, Weather, Electricity, and Traffic. This subset covers hourly ETT data, minute-level ETT data, meteorological variables, large-scale electricity load, and dense traffic flow. Table 7 reports the effect of removing or replacing the three CAPA components.

Table 7. Direct-test ablation study of positive-negative phase attention and dynamic fusion. Each variant changes one component at inference time; results are averaged across all prediction horizons. Lower values are better.

Design	Pos.	Neg.	Fusion	ETTh1		ETTm2		Weather		Electricity		Traffic	
				MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
CAPA	✓	✓	Dynamic	**0.400**	**0.416**	**0.259**	**0.320**	**0.224**	**0.260**	**0.167**	**0.258**	**0.391**	**0.254**
w/o positive branch	×	✓	Dynamic	0.429	0.438	0.275	0.335	0.241	0.274	0.176	0.269	0.413	0.265
w/o negative branch	✓	×	Dynamic	0.419	0.432	0.274	0.334	0.234	0.269	0.177	0.270	0.418	0.267
w/o dynamic gate	✓	✓	Fixed 0.5	0.411	0.426	0.267	0.328	0.231	0.266	0.172	0.265	0.404	0.261

Positive phase branch analysis. Removing the positive branch leaves only the negative phase branch and causes degradation in the averaged representative results. The averaged MSE increases from 0.400 to 0.429 on ETTh1, from 0.224 to 0.241 on Weather, and from 0.391 to 0.413 on Traffic. This supports the view that phase-aligned recurrence remains a core periodic signal: many future phases are best predicted by historical phase positions with similar within-period semantics.

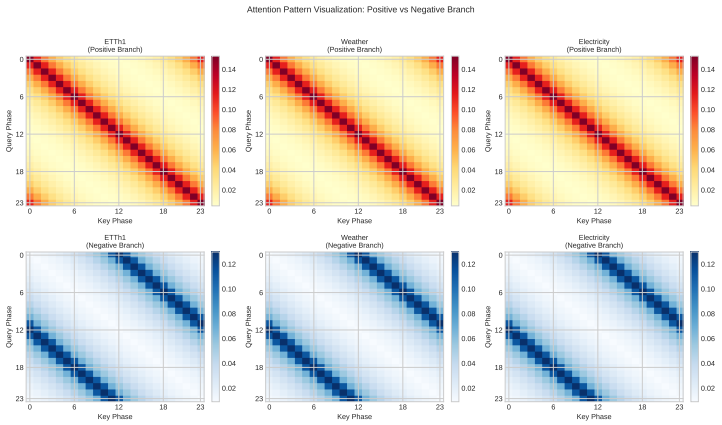


Fig. 4. Visualization of positive and negative phase attention. The two branches learn different phase-relation patterns: the positive branch places more weight on aligned or nearby phases, whereas the negative branch activates more offset phase candidates.

Negative phase branch analysis. When the negative branch is removed, the model can still use aligned phase attention, but it loses the explicit offset-biased view over phase positions. This variant is particularly worse on Electricity and Traffic, where peak–valley and high–low alternations are common. The result supports the usefulness of a complementary offset view for modeling phase interactions that are not captured by aligned recurrence alone.

Attention visualization. Fig. 4 visualizes the learned phase attention patterns. The positive branch concentrates more strongly around aligned phases, which matches its closeness-biased design. The negative branch distributes larger weights to offset phase positions, showing that it does not simply duplicate the positive branch. This visualization supports the interpretation that the two branches learn different phase-relation structures before dynamic fusion.

Dynamic fusion analysis. Replacing the learned fusion gate with fixed 0.5/0.5 averaging also degrades performance, although the drop is smaller than removing an entire branch. This behavior is expected: the fixed gate keeps both information sources but cannot adapt their relative importance for different samples, variables, and horizons. The dynamic gate therefore improves robustness by softly distinguishing windows dominated by aligned recurrence from windows better explained by offset phase interaction.

Ablation fairness. The ablation variants are not retrained smaller models. Each variant changes only one component: one branch output is masked, or the dynamic gate is replaced by fixed averaging. This protocol keeps the architecture and parameter budget aligned across variants, so the component comparison is not affected by parameter-count differences.

6.4 Analysis and Discussion

Across ETT, Weather, Electricity, and Traffic, CAPA benefits from folding long histories into phase tokens and learning two complementary phase-relation views.

The positive branch preserves aligned periodic recurrence, while the negative branch broadens the candidate dependency family toward offset phase positions. The dynamic gate further improves robustness by adapting the two branches for different samples, variables, and horizons. Consistently, the ablation results show that removing either branch or replacing the learned gate with fixed averaging degrades performance. The attention visualization in Fig. 4 also shows visibly different branch patterns, supporting the intended separation between aligned and offset phase interactions.

CAPA is intentionally compact: it changes the interaction domain from time positions to phase positions instead of adding a large sequence-level transformer stack. Its main assumption is that a meaningful primary period can be selected for each dataset. Following the experimental setting in Table 4, the period is obtained from frequency-domain analysis and remains fixed during training. For weakly periodic, rapidly drifting, or highly irregular data, future variants can combine multiple periods or add variable-level graph/routing modules to model cross-variable structure more explicitly.

7 Conclusion

We presented CAPA, a positive-negative phase attention model for long-term forecasting under periodic heterogeneity. The model reconstructs the sequence from a phase perspective, parameterizes aligned and offset-biased phase candidate views, and uses a dynamic gate to adapt their contributions. Main experiments across seven datasets and representative direct-test ablations show that the proposed components produce competitive forecasting accuracy while keeping the model compact. The current evidence supports the usefulness of the offset-biased branch, and the attention visualization further shows that the two branches learn different phase-relation patterns. Correlation-labeled diagnostics and broader evaluations remain important next steps.

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